

Tutorial for ESI and CWC Comparison for Abiotic Stressor-Drought

Since 2018, Europe has experienced persistent sub-seasonal-to-seasonal flash droughts, which are considered an unprecedented climatic phenomenon in the region over the past 250 years. Flash droughts are concerning due to their rapid onset and intensification by heatwaves and rainfall deficit. This leads to rapid soil moisture depletion, causing crops to desiccate and die faster than in slow droughts, especially during critical crop growth stages, which affects the yield. The early detection of flash droughts is possible through the evaluation of the response of plant biophysical variables to these events.

The summer 2022 and spring 2023 episodes were one of Europe's deadliest droughts on record, resulting in a staggering potential revenue loss of €25–30 billion for farming, affecting approximately half of the region, in which maize and wheat crops were notably affected.

The aim of this practical is to provide an introduction into Abiotic Stressor-Drought processing using Sentinel 2 images and the Sentinel Application Platform (SNAP). SNAP is an image processing package that can open many different types of satellite imagery and common GIS vector datasets. SNAP is open source and freely available for download from the European Space Agency website (<https://step.esa.int/main/download/>)

Will also be required Anaconda Navigator or Colab to process a time series of drought and non-drought years. You can find Anaconda application in the folder named Software.

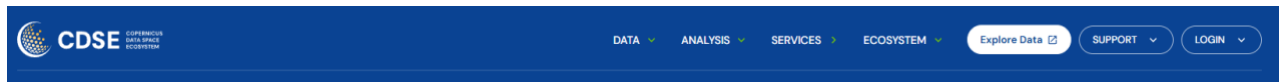
Reference:

Zaib Unnisa, Booker Ogutu, Jadunandan Dash. 2025. Potential of sentinel 2-derived canopy water content as an indicator of flash drought: Case studies from European cereal crop areas. *Remote Sensing Applications: Society and Environment*, Volume 39, 2025, 101690, ISSN 2352-9385. Available at: <https://doi.org/10.1016/j.rsase.2025.101690>

Parte 1: Copernicus Browser

1. Go to [Copernicus Sentinel Data | Copernicus Data Space Ecosystem](#)
2. Click 'Explore Data' then **Login**.

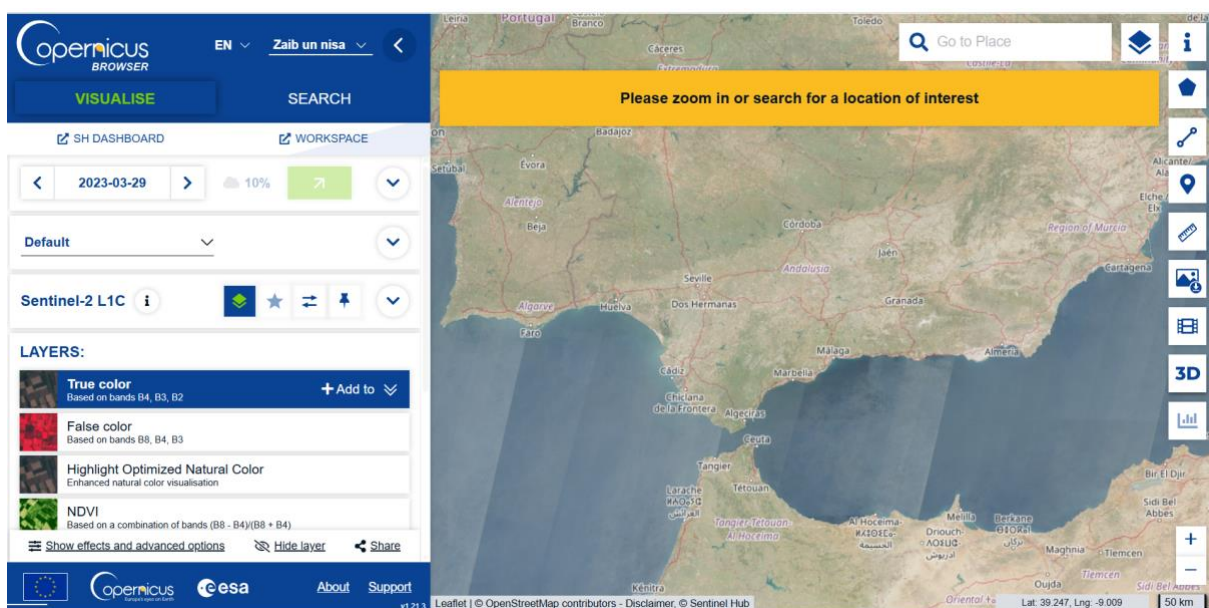
**For this exercise don't browse under 'Anonymously', it will not allow you to download data.*



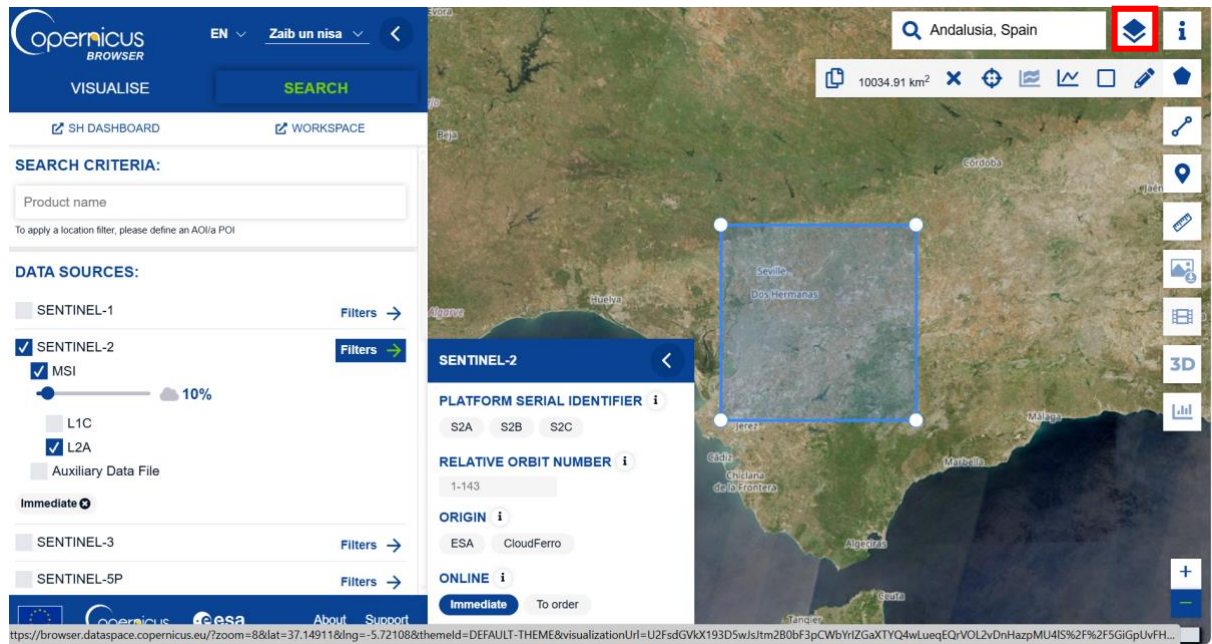
3. If you **have** an account, go to Login, then go to the Copernicus Browser Window

The image shows the login and registration interface of the Copernicus Browser. On the left, there is a 'Login to access your account' section with fields for Email and Password, a 'Forgot Password?' link, and a 'LOGIN' button. On the right, there is a 'Register and create an account for free in 60 seconds' section with three bullet points: 'Access a variety of Earth observation data', 'Manage your personal settings', and 'Follow your credits and orders'. Below these is a 'REGISTER' button.

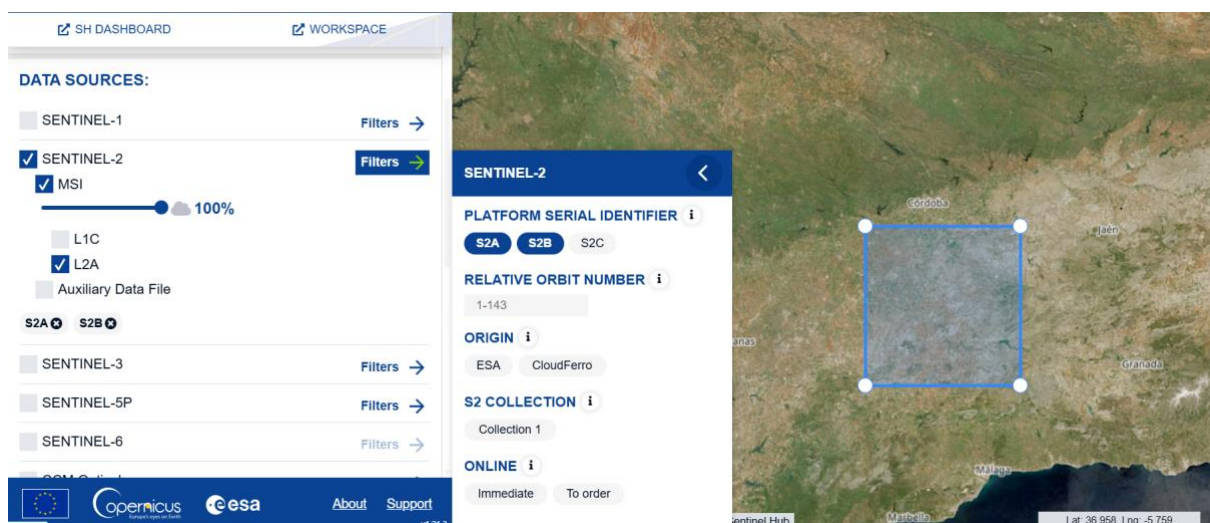
4. If you **don't have** an account, click on 'Register' and fill the form (you will receive a confirmation and an email validation to your email). With that, you will have access to download Sentinel 2 data.
5. After login, Explore Sentinel 2 data in the Copernicus Browser Window ([Copernicus Browser](#))
6. Zoom in to Andalusia, Spain (southern Spain).



7. Upload the SEV KML file or you can draw a rectangle over Seville using the drawing tool.



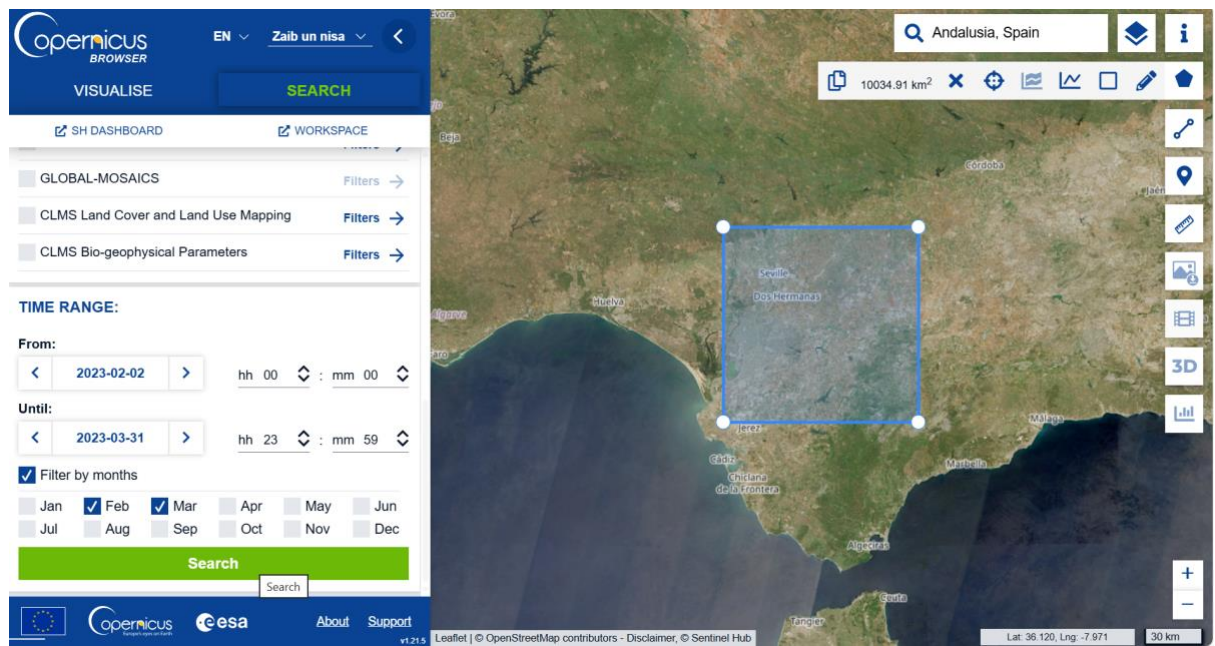
8. From Data Sources select 'SENTINEL-2 → MSI → L2A'.
9. Under MSI, you can filter the selection to a percentage of cloud cover, drag the filter to 20%



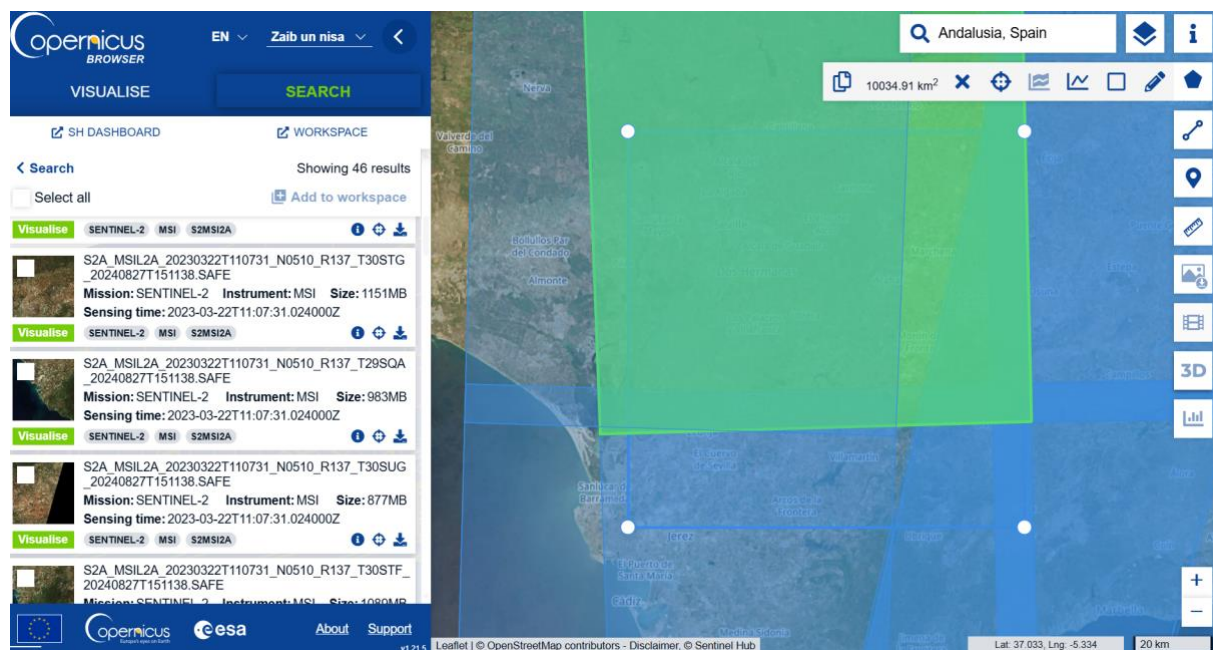
10. Under TIME RANGE, there are two sets of dates
 - a. For the drought year, filter the search FROM: 2023-03-01, UNTIL: 2023-03-31
 - b. For the non-drought year, filter the search FROM: 2018-03-01, UNTIL: 2018-03-31

**You can also select any other date based on your ESI timeseries.*

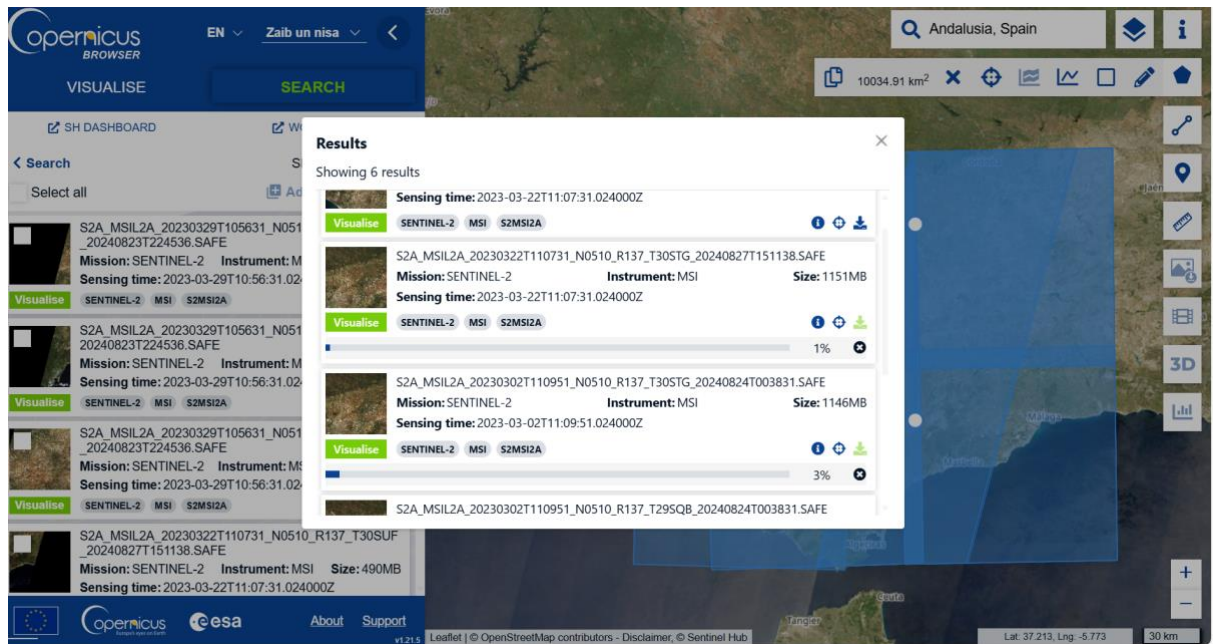
11. Download 4 images, for example:



12. Select the image that maximally covers the KML area.



13. Click on the Download icon for the first image. Save in a known location.



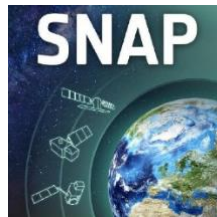
14. Repeat this exercise for other dates and the same view by going back to the search window. You should have this list of products:

- S2A_MSIL2A_20180328T110741_N0500_R137_T30STG_20230828T152417.SAFE
- S2B_MSIL2A_20180221T111049_N0500_R137_T30STG_20230912T113537.SAFE
- S2A_MSIL2A_20230302T110951_N0510_R137_T30STG_20240824T003831.SAFE
- S2A_MSIL2A_20230322T110731_N0510_R137_T30STG_20240827T151138.SAFE

**Track the names, because the original files are renamed during unzipping.*

Parte 2: SNAP

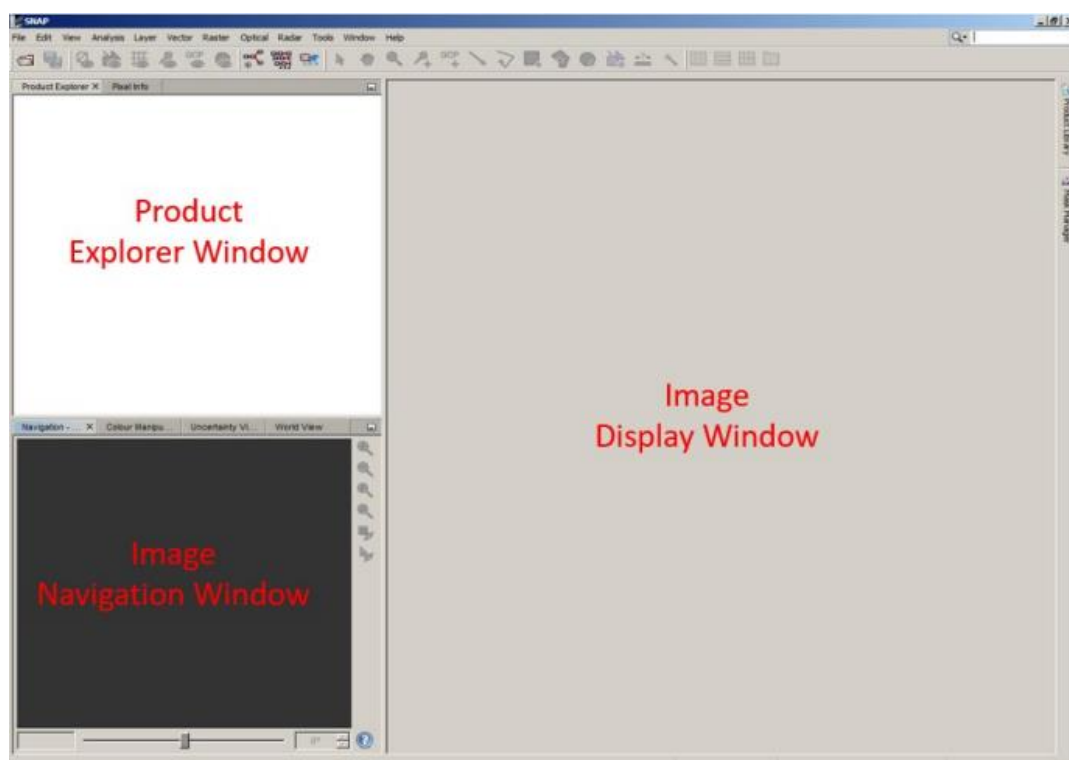
1. Install SNAP10 from the Science Toolbox Exploitation Platform ([SNAP Download – STEP](#))
2. Download the correct version of the **Sentinel Toolboxes** depending on your operating system. Follow the instruction in the download windows and restart your machine if necessary.
3. To launch SNAP, double click the left-mouse button on the SNAP icon on the desktop:



This will launch the SNAP software which has a graphical user interface (GUI) that contains a series of menus and buttons at the top of the screen (figure below).

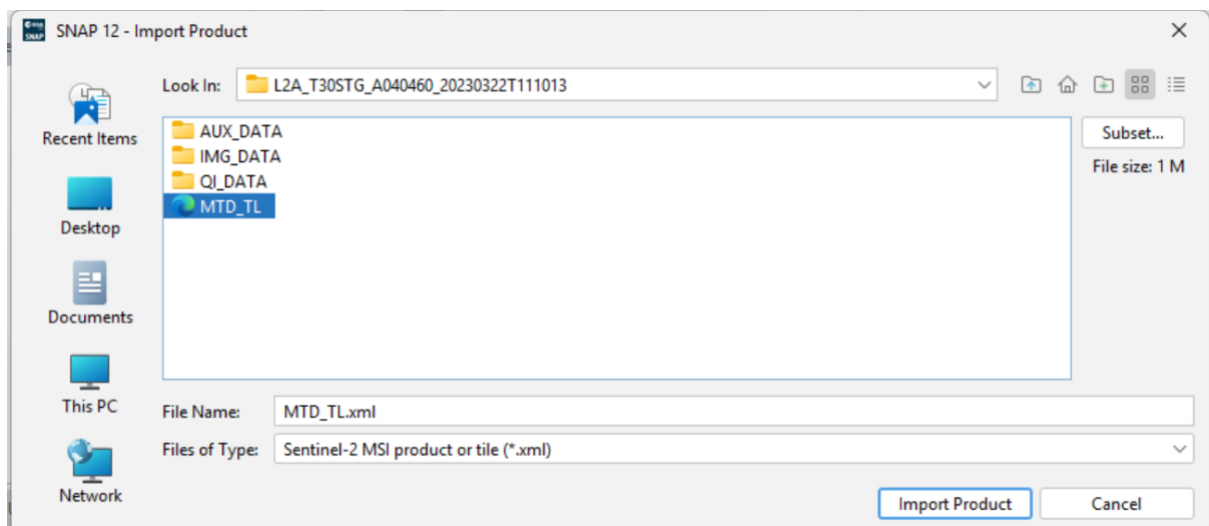
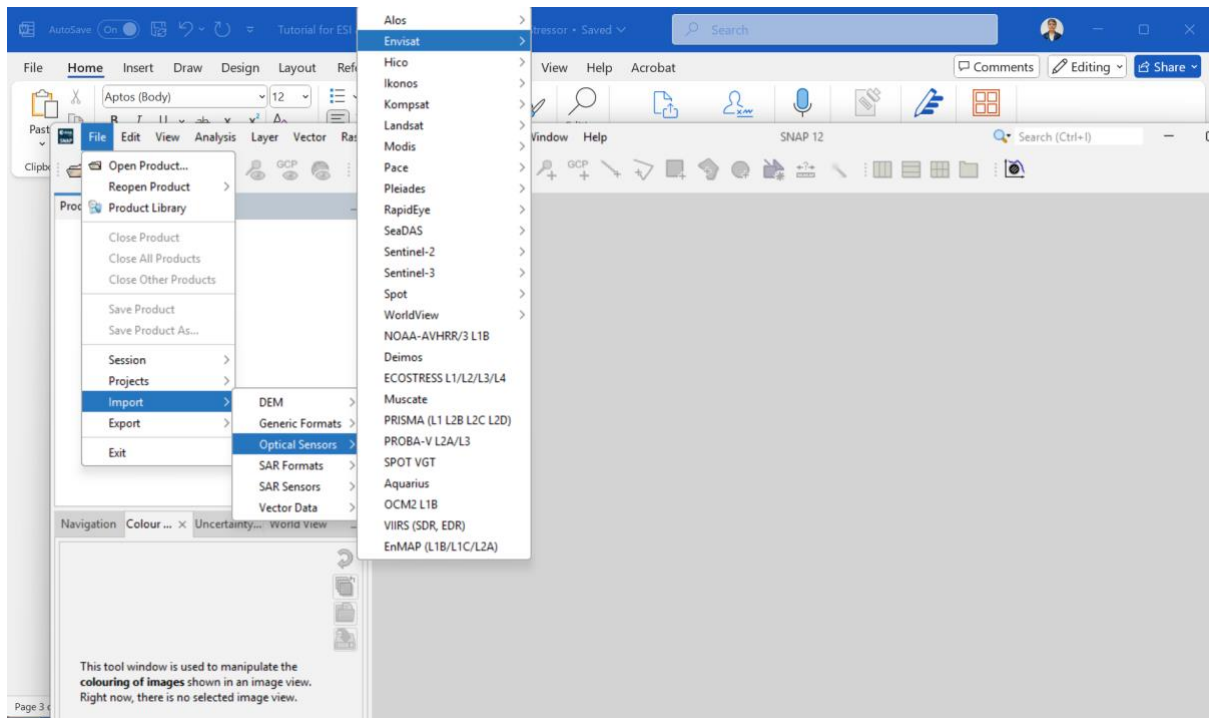
The interface contains three windows.

- **Product Explorer Window** -> the upper left window provides information on the image that has been loaded such as the image bands and pixel values.
- **Image Navigation Window** -> the lower left window provides information on the image values and image navigation information such as zooming in and out of the displayed image.
- **Image Display Window** -> the main image display window displays the image that has been loaded



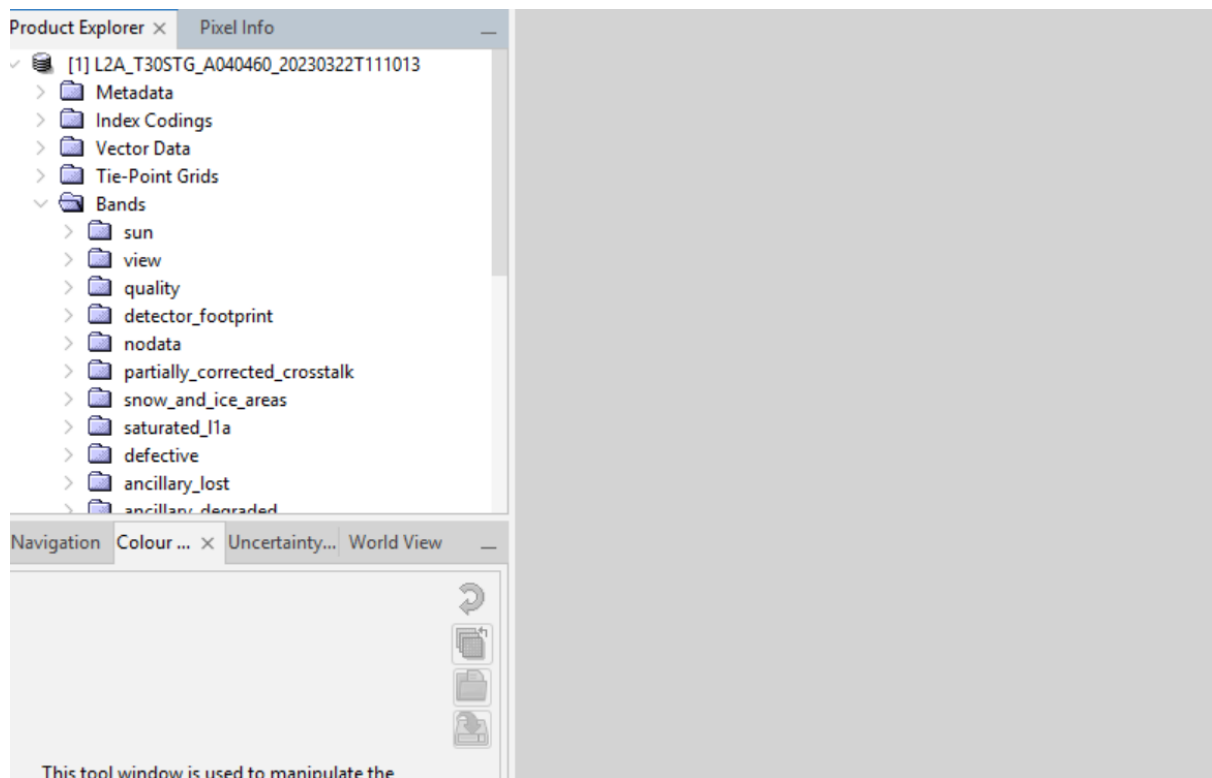
4. **Import** the images downloaded from Copernicus to SNAP Software.

5. To load an image, press the **open icon**  in the upper left-hand corner of the window. Navigate to the directory and folder that contains the satellite products (images) that you downloaded in the previous exercise. Select the **.xml** file to open the images in the software.

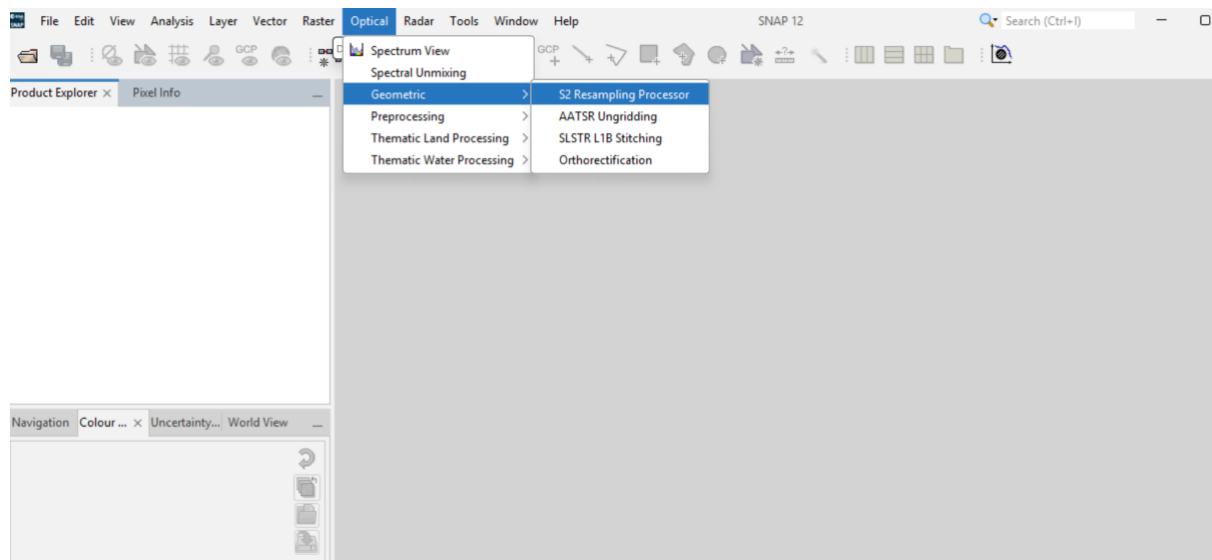


6. The images will then be opened with the full name of the downloaded product and will be shown in the **product explorer window** (top left window).

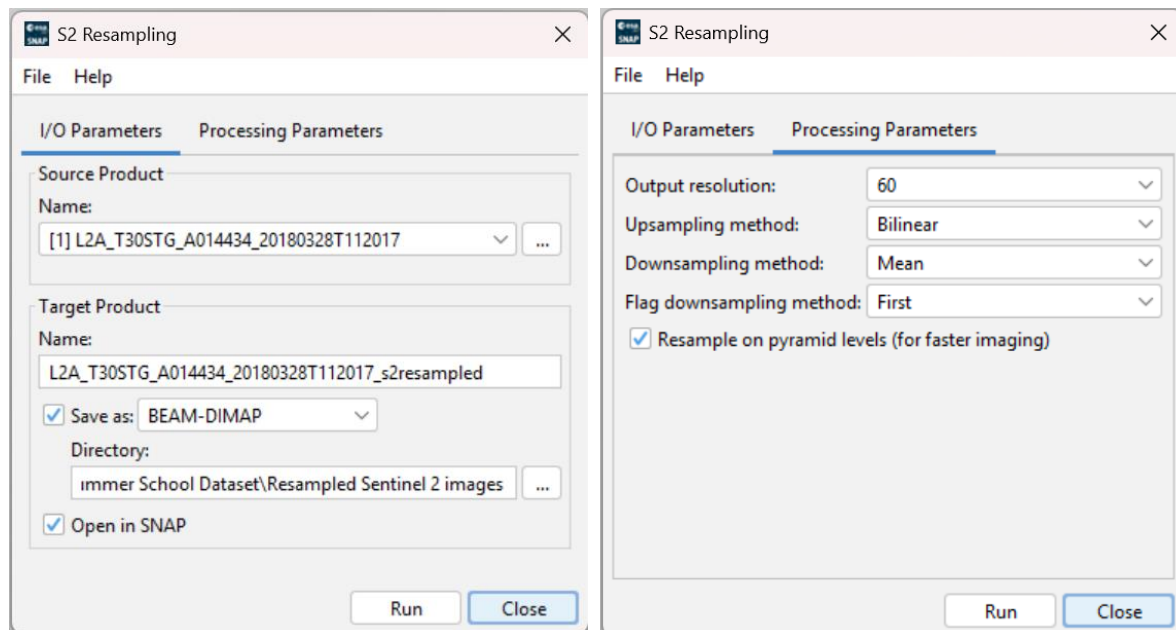
7. **Press +** next to the products name to show various directories associated with the images (such as 'Metadata', 'Vector Data', 'Bands', etc)



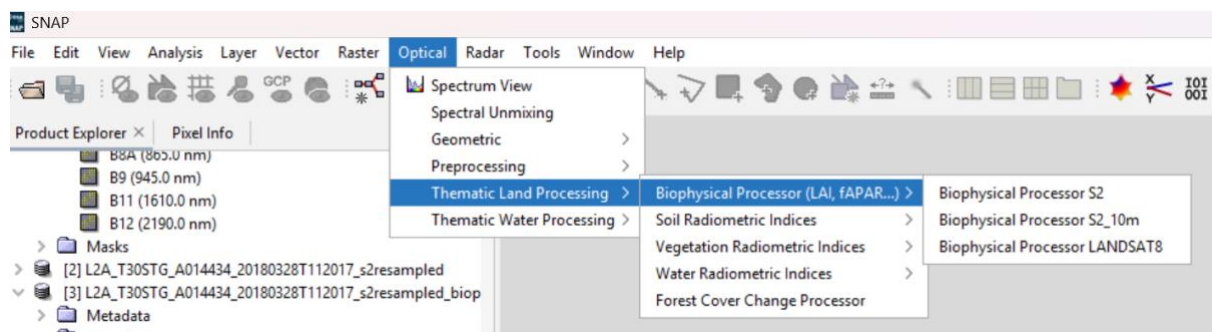
8. **Explore** uploaded bands (double click for display; bottom right corner to explore properties and data)
9. We need to resample Band 8 from each image, it has a 10 m resolution, and it need either 60 m or 30 m resolution to be comparable. **Go to Optical → Geometric → S2 Resampling Processor**



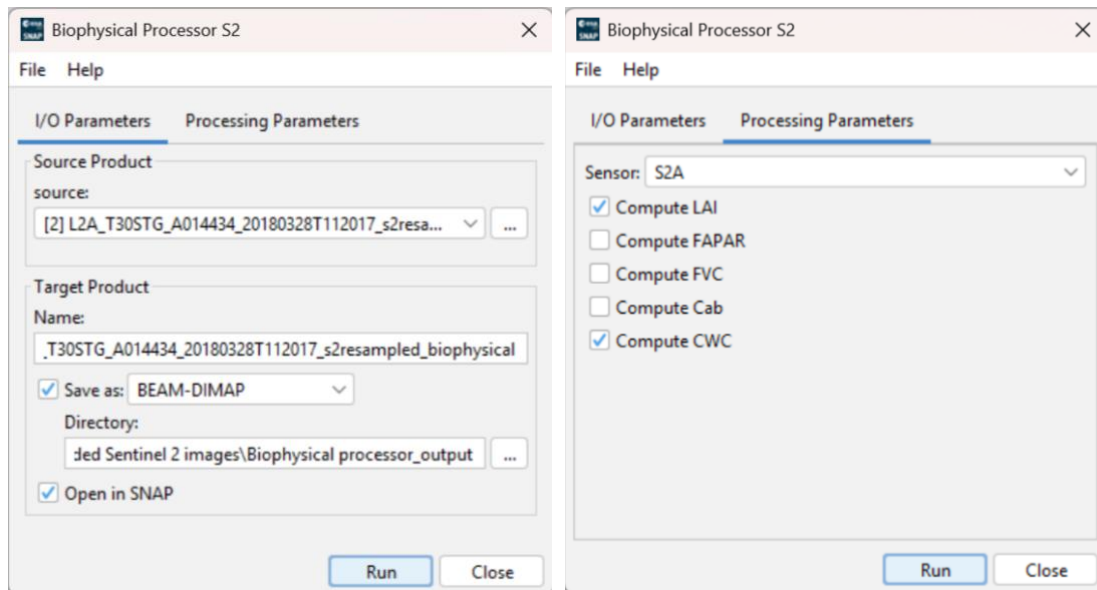
10. In Directory (I/O Parameters → Target Product) create a new folder to contain the resulting resampling products.
11. In the Processing Parameters tab, adjust parameters for resampling using the details in the image below
 *you can explore different output resolutions (10, 20, 30m, etc), it might take longer to process.



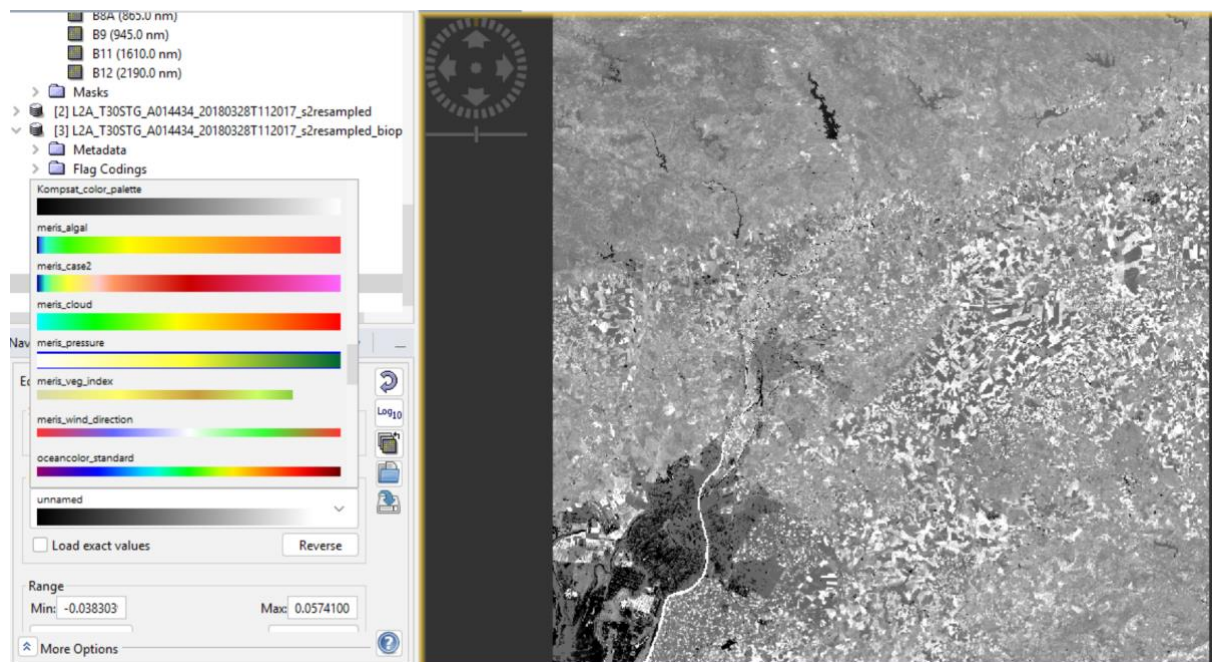
12. Go to Optical again select → **Thematic Land processing** → **Biophysical Processor** → **Biophysical Processor S2**



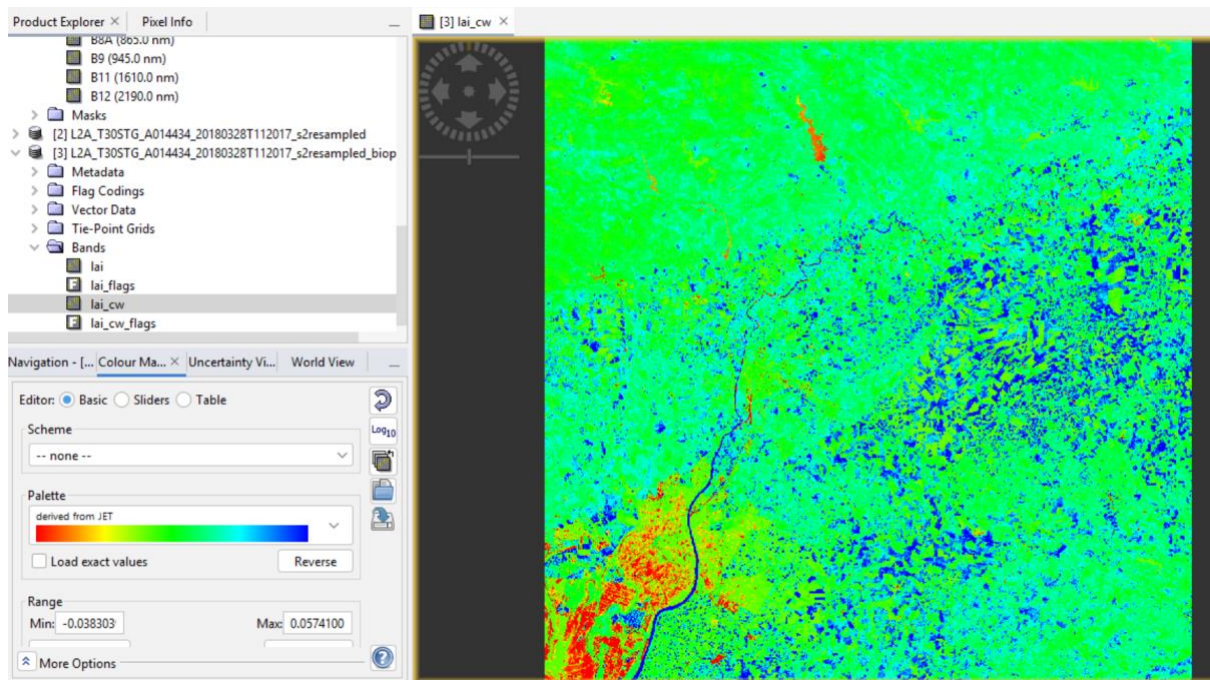
13. Adjust I/O and processing parameters using the details in the image below (make sure is the _s2resample result image)



15. Choose sensor type S2A or S2B based on data file. Choose only LAI and CWC for this exercise; you can select all variables if interested.
16. You have available now an image with biophysical variables bands
17. Display the CWC band and change colour scheme in the 'Colour Manipulation' tab (bottom left corner).

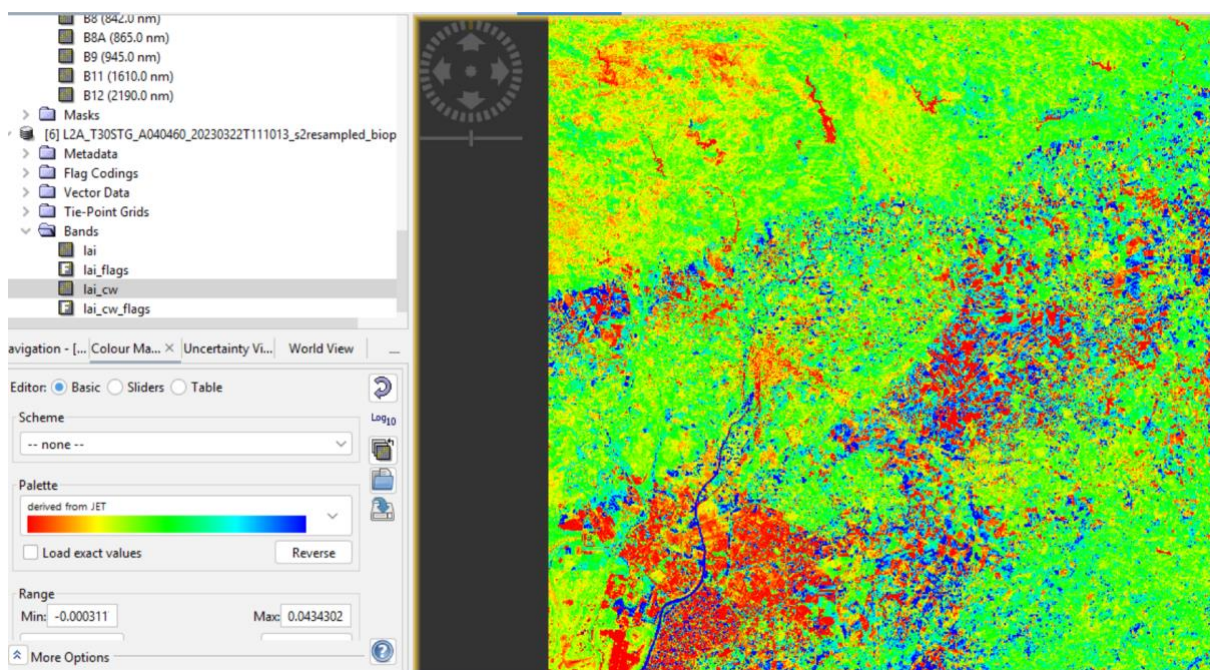


18. Change the default palette Scheme from 'unnamed' to 'Jet'. The image below shows the CWC pattern for 28 March 2018 for different crops.



19. Repeat steps 1 – 18 for other images.

20. The image below shows the CWC pattern for 22 March 2023. You can see how the CWC pattern changed in the drought year.



21. Now, plot the same pattern for LAI with a different colour scheme and discuss the impact of drought on agriculture.